Description

Illustration



Figure caption

Concretes designated as 'high performance' have been used in some of the most iconic modern buildings. 1. Petronas Towers, Kuala Lumpur. 2. The piers of the Millau Viaduct, France are made of high performance concrete. 3. The tallest building in the world, the Burj Khalifa in Dubai. Images used under license from Shutterstock.com

The material

High performance concrete (HPC) is generally characterized by the following properties; ease of placement, compaction without segregation, early age strength, long-term mechanical properties, permeability, high density, toughness, volume stability and long-life in severe environments generally due to the lack of connecting capillary pores. HP cement and mortars with compressive strength up to 95 and 145 MPa, respectively can be produced. To achieve these properties the formulation of HPC is carefully regulated to maximize compaction and homogeneous distribution of all constituent materials in the mix. High performance concrete has generally replaced high strength concrete as the designation of the array of concrete mixes that achieve performance well beyond that of normal concrete. Note that the properties here regard unreinforced concrete, rather than the significantly improved properties that can be achieved by reinforcement with concrete.

Compositional summary

139:513:43:685:1080 Water:OPC:Silica Fume:Fine:Coarse (119MPa Fcu)

General properties

Density		2.2e3	-	2.6e3	kg/m^3
Price	*	0.0604	-	0.128	GBP/kg

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓

Enclosure	✓			
Interiors	✗			
Services	✗			

Mechanical properties

Young's modulus	32	-	43	GPa
Shear modulus	* 13.9	-	18.7	GPa
Bulk modulus	* 15.2	-	20.5	GPa
Bending modulus	* 32	-	43	GPa
Poisson's ratio	* 0.1	-	0.2	
Yield strength (elastic limit)	* 5.3	-	9.3	MPa
Tensile strength	* 5.3	-	9.3	MPa
Compressive strength	* 53.3	-	93.3	MPa
Bending strength	* 6.4	-	11.2	MPa
Elongation	* 0.01	-	0.03	% strain
Hardness - Vickers	* 20.1	-	22.2	HV
Fatigue strength at 10^7 cycles	* 3.15	-	4.94	MPa
Fracture toughness	0.35	-	0.45	MPa.m^0.5
Mechanical loss coefficient (tan delta)	* 0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?	Poor insulator			
Thermal resistivity	* 0.385	-	0.606	m.°C/W
Thermal expansion coefficient	* 7	-	12	µstrain/°C
Specific heat capacity	1e3	-	1.18e3	J/kg.°C
Melting point	* 927	-	1.2e3	°C
Maximum service temperature	* 627	-	857	°C
Minimum service temperature	-160	-	-150	°C
Flammability	Non-flammable			
Emissivity	0.63	-	0.97	

Hygro-thermal properties

Water vapor permeability	2e-12	-	6.01e-12	kg.m/s.m^2.Pa
Frost resistance	Average			

Electrical properties

Electrical conductor or insulator?	Poor insulator			
Electrical resistivity	* 1.85e12	-	1.85e13	µohm.cm
Dielectric constant (relative permittivity)	* 8	-	12	

Dissipation factor (dielectric loss tangent)	* 0.001	-	0.01	

Optical properties

Natural color

Gray

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No			

Acoustic properties

Sound absorption	Poor			
Sound isolation	Good			

Processability (scale 1 = impractical to 5 = excellent)

Castability	5			
Moldability	3	-	5	

Durability

Water (fresh)	Excellent			
Water (salt)	Excellent			
Weak acids	Acceptable			
Strong acids	Unacceptable			
Weak alkalis	Excellent			
Strong alkalis	Acceptable			
Organic solvents	Excellent			
UV radiation (sunlight)	Excellent			
Industrial atmosphere	Excellent			
Rural atmosphere	Excellent			
Marine atmosphere	Acceptable			

Geo-economic data for principal component

Annual world production, principal component	1.43e10	-	1.58e10	tonne/yr

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.16	-	0.177	kg/kg
Embodied energy (virgin grade production)	* 1.17	-	1.3	MJ/kg
Water Usage	* 2.86	-	3.17	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 21.7	-	27.7	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.42	-	1.81	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.016	-	0.0177	kg/kg
Embodied energy, recycling	* 0.164	-	0.182	MJ/kg
Recycle fraction in current supply	14			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

High Performance (HP) concrete is a modified concrete formulation based on special cements and admixtures, achieving elevated compressive strengths.

The production of HP is specialist, but achievable by major ready-mixed concrete suppliers. The strength range brings it within the scope of normal reinforced concrete structural design methods.

Application of HP concrete reduces the expense of construction, with the most important economic benefits being:
- energy saving due to reduced cross sections of structures
- reduced construction cost due to less total weight of structures and foundations, reduced total energy consumption, less formwork, less maintenance and equipment loading, increased build speed
- significantly increased life-time of structures and reduced cost of repair work due to high durability (American Concrete Institute)

Technical notes

The HP cement technology is based on:
- modification of Portland cement by complex admixtures
- incorporation of ground granulated blast-furnace slag and silica fume into the cement composition
- optimization of mixture design
- mixing and grinding process optimization.

HP cement technology is based on mechanical-and-chemical activation of the cement grinding process for increasing dispersion and reaction ability, as well as modifying of material surface by chemical admixtures. The three main processes that take place at HP cement production are:
- pulverization of clinker and slag due to development of micro-distraction
- creation of high active amorphous structures
- physical and chemical modification of the cement surface by complex admixtures.

Note that the properties here regard unreinforced concrete, rather than the significantly improved properties that can be achieved by reinforcement with steel bar and fibres.

Typical uses

HP cement-based concrete is used increasingly in high-rise buildings, airport runway structures, bridges, marine and offshore structures, tunnels, parking decks, underwater and marine structures, special floors, high performance vessels and containers.

Tradenames

Duracrete (Hanson)

Related standards

The properties of Cement and Concrete vary a great deal based on composition, age and processing. Standards exist to help ensure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main US, EU and UK standards related to this material. Other countries have their own standards, for example: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

AS 3972: General purpose and blended cements
ASTM C150: Standard Specification for Portland Cement
ASTM C1697: Standard Specification for Blended Supplementary Cementitious Materials
EN 197-1: Cement. Composition, specifications and conformity criteria for common cements
EN 206: Concrete. Specification, performance, production and conformity
BS 8500: Concrete. Complementary British Standard to BS EN 206
CSI/CSC 03 31 23: High-Performance Structural Concrete

Further reading

ACI Committee 363, "State-of-the-Art Report on High-Strength Concrete," ACI Manual of Concrete Practice, Vol.1, American Concrete Institute, Detroit, 1993 and later editions, pp. 363R-1 to 363R-55.

Links

Producers

Reference